

Tr	ID	User Story	🔗 MVP / Reach	Function	Task	🔗 Owner	🔗 Priority	Est.	Actual	🔗 Status	Acceptance Criteria
1.1		As a researcher, I want to use and improve Google's neural-network quantum decoder solution, so that I can reduce errors in a quantum computer better than other state-of-the-art decoder solutions.	MVP	Set Up	Set up github repository under which out decoder solution lives	Christopher	High	0.75	3	Done	Accessible github repository in which everyone can run and edit simple Readme file
1.2			MVP		Set up virtual environment for decoder solution	Christopher	High	1.5	3	Done	Only when environment is active does file inside run
1.3			MVP		Create Docker image and container for decoder	Christopher	High	1	3	Done	Verify image and container exists using docker ps or docker container ls
1.4			MVP		Run test Docker containing virtualizations	Christopher	Low	0.75	1	Done	Check if helloworld python file runs only after running container
1.5			MVP		Study optimizer LION from Google's paper	Tzu Chen	Medium	4	13	Done	Understand what the optimizations implemented by the Google paper are
1.6			MVP		Study data from Google's paper	Tzu Chen	Medium	10	12	Done	Understand the data used and implemented by Google's paper
1.7			MVP		Installing and learning ARC	Tzu Chen	Medium	3.5	3	Done	Run example python file in the ARC successfully and get output results
1.8			MVP		Study transformer and neural network models	Mara	Medium	5	4.5	In Progress	Understand neural network and transformer models to the required extent
1.9			MVP		Study AlphaQubit papers	Arjun	Medium	5	8	In Progress	Understand the AlphaQubit implementation and performance
1.10			MVP	Run on Simulated Dataset	Create surface code code in Stim to simulate surface code error dataset	Christopher	Medium	4.5	4	Done	Stim circuits created for $d \in \{3,5\}$ and specified noise model(s).
1.11			MVP		Generate / simulate Stim datasets	Mara	Medium	3.75	2	Done	Script generates train/val/test splits with configurable shots and p values.
1.12			MVP		Implement analog readout wrapper	Mara	Low	3	4	Done	Wrapper accepts soft information (e.g., logits/LLRs/probabilities) and returns calibrated analog values.
1.13			MVP		Connect to the ARC	Arjun	High	3.75	3	Done	SSH key + MFA working; sinfo & sbatch accessible. SLURM template committed (e.g., slurm/train_surface.sh)
1.14			MVP		Construct embedding scheme (with vectorizing training data and tokenization)	Arjun	Medium	7	1	Done	Deterministic mapping from syndrome/analog readout to tokens/embeddings.
1.15			MVP		Implement proposed transformer error correction architecture with pipeline	Christopher	Medium	5.25	6	Done	Model modules implemented with clear interfaces and docstrings. End-to-end train.py and eval.py accept con
1.16			MVP		Train on Stim data on ARC	Arjun	Medium	4.75	6	In Progress	SLURM job(s) complete without error; logs and checkpoints saved.
1.17			MVP		Test and evaluate performance on Stim data	Mara	Medium	2.5	4	Done	eval.py runs across held-out splits and p-grid; outputs LER(p) table.
1.18			MVP		Iteratively adjust parameters and model architecture to improve performance on Stim data	Tzu Chen	Medium	34	30	In Progress	At least N (e.g., 5) controlled experiments executed (documented changes only). Ablation table added to repo
1.19			MVP		Preparing pretrain, valuation and test dataset	Tzu Chen	Medium	20	32	Done	
1.20			MVP	Run on Sycamore Dataset from Paper	Adjust embedding scheme / pipeline for sycamore dataset	Christopher	High	70	70	In Progress	Loader for Sycamore data implemented; normalization/calibration documented.
1.21			MVP		Train on Sycamore data on ARC	Arjun	Medium	5.25		Stalled	SLURM job completes. Checkpoints/logs archived separately from Stim runs.
1.22			MVP		Test and evaluate performance on Sycamore data	Tzu Chen	Medium	2.75		Stalled	LER(p) (or per-round logical failure) computed on held-out Sycamore set.
1.23			MVP		Iteratively adjust parameters and model architecture to improve performance on Sycamore data	Tzu Chen	Medium	3.75		Stalled	At least N (e.g., 5) controlled experiments executed for Sycamore dataset specific challenges (documented c
2.1		As a researcher, I want to apply Google's neural-network quantum decoder to other quantum codes, so that I have a generalized decoder applicable to any type of quantum code / system.	Reach	Run on Simulated Dataset	Create code other quantum codes using Stim	Mara	Low	7.25		Stalled	At least two additional codes (e.g., color code, XZZX surface) implemented.
2.2			Reach		Generate dataset for other quantum codes	Christopher	Low	4.5		Stalled	Datasets built with the same schema; metadata includes code_family.
2.3			Reach		Adapt readout wrapper to accept other datasets	Christopher	Low	4.25		Stalled	Wrapper adapts to code family differences via config (no code changes).
2.4			Reach		Adapt embedding scheme to accept different quantum code data (with unique vectorization schemes)	Arjun	Low	6.25	8	Done	Single embedding API supports all code families (feature flags). Validation asserts consistent tensor shapes
2.5			Reach		Train on Stim data for other quantum codes	Christopher	Low	5.25		Stalled	Training runs complete for each additional code (min one config each).
2.6			Reach		Test and evaluate performance on Stim data for other quantum codes	Christopher	Low	5.5		Stalled	LER(p) curves produced and saved for each code.
3.1		As a graduate student, I would like to compare my decoder with a pre-built transformer-based decoder and baseline models	MVP	Implement baselines	Implement MWPM baseline error correction on simulated Stim data	Arjun	Medium	3.75	2	Done	MWPM decoder (e.g., PyMatching) integrated with same data interface.
3.2			MVP		Evaluate MWPM baseline performance on Stim data	Tzu Chen	Medium	2.5	2	Done	LER(p) for MWPM on surface code computed and saved.
3.3			MVP		Implement MWPM baseline error correction on Sycamore data	Tzu Chen	Medium	4		Stalled	Sycamore loader into MWPM path runs end-to-end.
3.4			MVP		Evaluate MWPM baseline performance on Sycamore data	Mara	High	3.5		Stalled	LER results computed; plot and results exported.
3.5			MVP	Implement Google's transformer model	Build branch from neural net pipeline to send embedded data to Google's transformer model	Tzu Chen	Low	4.25	1	Done	Separate training path -model google_transformer implemented.
3.6			MVP		Build transformer based on Google's pseudo code	Tzu Chen	Medium	8	10	Done	Verify it utilizes same parameters, architectural design, is able to handle similar error data, and performs at le
3.7			MVP		Run transformer based on Google's pseudo code over ARC	Arjun	Medium	5	8	Done	Verify a run can be made over the provided allocation over ARC using STIM generated data and model based c
3.8			MVP		Train Google's transformer model on Stim Data	Christopher	Medium	5	12	In Progress	Training completes with logged metrics and checkpoints.
3.9			MVP		Evaluate Google's transformer performance on Stim data	Tzu Chen	Low	2.75		To Do	LER(p) computed and saved. Plots exported.
3.10			MVP		Train Google's transformer model on Sycamore Data	Arjun	Medium	5		Stalled	SLURM job completes. Artifacts saved.
3.11			MVP		Evaluate Google's transformer performance on Sycamore data	Christopher	High	3.25		Stalled	Evaluation artifacts produced. Export plots and results.
3.12			MVP	Evaluation Module	Compare performance with proposed solution using LER (p) evaluation metric	Arjun	High	3.5		To Do	- Unified comparison plot with all decoders (ours, Google's, MWPM) for Stim & Sycamore. - Table reports LER at each p, relative improvement vs MWPM, and crossover points. - AUC (LER vs p) computed; bootstrap CIs provided. - Narrative summary added to report.
3.13			MVP		Generate visualizations and report	Mara	Medium	30	29.5	In Progress	Figures: LER(p), training curves, ablations, throughput/latency, calibration.
4.1		As a graduate student, I want to integrate a neural-network quantum decoder solution into a simplified quantum model.	Reach	Decoder Adoption	Adapt readout module	Arjun	Low	2.75		Stalled	Streaming input (e.g., ZeroMQ/Kafka/pipes) interface implemented.
4.2			Reach		Feed into pipeline of transformer model that evaluates error correction output	Tzu Chen	Low	3.5	6	Done	End-to-end streaming decode runs continuously >30 minutes without crash.
4.3			Reach		Evaluate and verify real-time performance and throughput	Tzu Chen	Low	4		Stalled	Latency (p50/p95) and throughput measured at batch=1 and batch=256.
4.4			Reach		Research adapting decoder to other quantum code types	Arjun	Low	10	6	In Progress	Finalize possible adaptation documentation with steps to be taken, concerns, and future work
4.5			MVP	Interface and packaging	Construct interface to provide option to modify parameters	Mara	Medium	4.25		Stalled	CLI + minimal UI (e.g., Streamlit or REST) exposes key params (depth, heads, lr, batch, calibration). Validation
4.6			MVP		Build documentation and details on model functionality and	Arjun	Medium	12	8	In Progress	Verify correctness with Alphaqubit paper and ensure model data and details are accurate